Case Study 8: Even or Odd Number Checker

1. Problem Statement

Read an integer from the user and check whether it is even or odd.

2. Objectives

1. Read an integer.
2. Use the modulus operator.
3. Apply an if-else condition.
4. Display whether the number is even or odd.

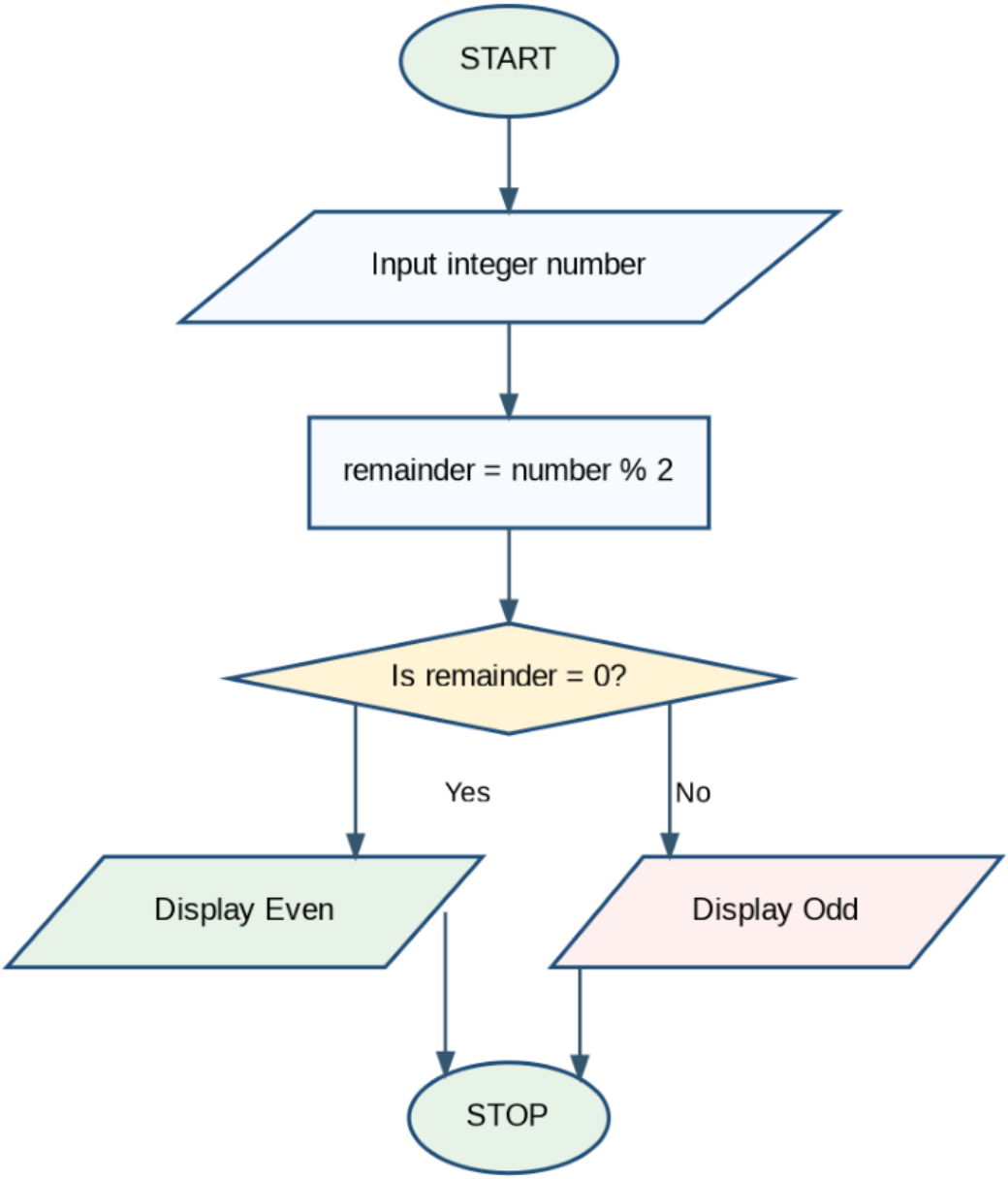
3. Assumptions

1. The user enters a valid integer.
2. A number is even when its remainder after division by 2 is zero.

4. Algorithm

1. Start.
2. Read an integer.
3. Calculate remainder = number % 2.
4. If remainder is zero, display Even.
5. Otherwise display Odd.
6. Stop.

5. Flowchart



6. Pseudocode

```
INPUT number
IF number % 2 == 0 THEN
    DISPLAY "Even"
ELSE
    DISPLAY "Odd"
```

7. Complete Python Source Code

```
number = int(input("Enter an integer: "))
if number % 2 == 0:
    print(number, "is Even")
else:
    print(number, "is Odd")
```

8. Test Case and Expected Output

Input = 17.

17 % 2 = 1.

Expected output: 17 is Odd.

9. Conclusion

The program checks whether a number is even or odd using the modulus operator and if-else.